

Rohan Kallur

rkallur135@gmail.com | 732-570-0627 | rohankallur.xyz | github.com/RKPurple | linkedin.com/in/rohankallur

Education

Stevens Institute of Technology

Sept 2021 – May 2025

Bachelor of Science in Computer Science, Minor in Finance

Hoboken, NJ

- GPA: 3.7/4.0 | *Magna Cum Laude*
- Relevant Coursework: Data Structures, Algorithms, Object-Oriented Development, Agile Methods, Database Management Systems, Mobile Systems and Applications, Intro to Quantitative Finance, Financial Accounting, Corporate Finance, Intro to Derivatives
- Academic Honors: Dean's List (All Semesters), Upsilon Pi Epsilon (Computer Science Honor Society)

Skills

Languages: TypeScript, Python, JavaScript, Java, SQL, Solidity, C/C++, Rust

Tools and Frameworks: React, Node.js, PostgreSQL, FastAPI, Docker, AWS, Django, Git, Unix, PyTorch, TensorFlow, OpenAI, Playwright, Jupyter Notebook, Google Colab, React Native, Web3.js, Atlassian, Roboflow, Weights & Biases

Work Experience

Software Engineer

May 2026 – Present

AvaWatz

Remote

- Building a clinical AI tool to help dentists detect teeth-positioning shifts, owning work across the full ML pipeline.
- Developed a YOLO-based instance segmentation pipeline in Google Colab to generate and correct teeth masks across 150 real patient images (Roboflow), then fine-tuned the model in PyTorch with experiments tracked in Weights & Biases.
- Engineering an automated TMJ condylar-position analysis pipeline in Python (NumPy, OpenCV, PIL) deriving diagnostic grids from YOLO segmentation masks on ~140 panoramic radiographs, with tunable grid-construction variants refined through feedback from practicing dentists.

Software Development Intern

May 2024 – Aug 2024

Netki

Remote

- Developed an Ethereum smart contract (Solidity) for DeFi token verification, enforcing KYC compliance for digital assets; used Truffle and Ganache for local blockchain simulation.
- Refactored smart contract logic to reduce gas costs by 50%, optimizing event emissions, storage layout, and algorithm logic to cut unnecessary computation.
- Built automated test suites with Mocha/Chai, enabling reliable CI/CD integration and confident production deployments.
- Designed system architecture and API documentation for open-source release.

Research Assistant

Jan 2023 – May 2023

Stevens Institute for Artificial Intelligence

Hoboken, NJ

- Built Python prototypes to explore NLP (Natural Language Processing) and Generative AI pipelines as part of an NSF-funded collaboration with Columbia and Syracuse — focused on extracting structured information from unstructured data sources.
- Utilized OpenAI GPT-4 (beta) to generate headlines from image datasets using vector embeddings, similarity metrics, and clustering algorithms.
- Used PyTorch to fine-tune models and build interactive tools for computational journalism applications.

Projects

NBA ReDrafter *redraft-room.onrender.com*

Jan 2026 – Present

- Designed and shipped a full-stack web app end-to-end using React 19, TypeScript, FastAPI, and PostgreSQL — including drag-and-drop UI, shareable links, and screenshot export via Playwright headless browser automation.
- Engineered a Python data pipeline to scrape, merge, and ingest historical NBA draft data into PostgreSQL, covering 25 draft classes (2000–2025) and 2,500+ players.
- Architected and deployed a REST API (FastAPI) with endpoints for data retrieval, image proxying, and token-based ephemeral storage — optimized for low-latency responses and stable uptime under concurrent load in production on Render.

Portfolio Website *rohankallur.xyz*

Feb 2026 – Present

- Designed, built, and deployed a personal website using Next.js 16, React 19, TypeScript and Tailwind CSS v4; built a CS:GO inspired case-unboxing entry animation using Framer Motion that reveals the content of the site.
- Included contact form backed by Next.js API route and Resend, with client-side validation and a honeypot for spam prevention. As well as light and dark theme switching using custom React context.